Contents

Exercises are from Stephen Abbott's *Understanding Analysis*, Second Edition.

I'll state some theorems here from the book that don't have names:

Theorem 1: 6.5.1 If a power series $\sum_{n=1}^{\infty} a_n x^n$ converges at some point $x_0 \in \mathbb{R}$, then it converges absolutely for any x satisfying $|x| < |x_0|$.

Theorem 2: 6.5.5 If a power series converges pointwise on the set $A \subseteq \mathbb{R}$, then it converges uniformly on any compact set $K \subseteq A$. As a consequence, it is continuous at every point at which it converges.

Problem 1: An alternating power series

Exercise 6.5.1

Define the function g by the power series

$$g(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

Let

$$a_n = (-1)^{n+1} \frac{x^n}{n}.$$

(a)

Let $x \in (-1, 1)$. Then $|x| < 1$ and

$$\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \to \infty} |x| \left| \frac{n}{n+1} \right| = |x| < 1,$$

so by the Ratio Test the series converges pointwise when $|x| < 1$. By Theorem 6.5.5 it converges uniformly on compact subsets of $(-1, 1)$ and is continuous on $(-1, 1)$.

At $x = 1$ the series converges by the Alternating Series Test, since

$$1 > \frac{1}{2} > \cdots > \frac{1}{n} > \cdots$$

and $\left(\frac{1}{n}\right) \to 0$. So the power series defining g is defined on $(-1, 1]$. By Theorem 6.5.5 it converges uniformly on every compact subset of $(-1, 1]$ and it's continuous on $(-1, 1]$.

At $x = -1$ this is (the negative of) the harmonic series, so it diverges (by the Cauchy Condensation Test). So g is not continuous on $[-1, 1]$.

The series cannot converge for any point x with $|x| > 1$ because that would imply convergence for all y with $|y| < |x|$, including $y = -1$, contradicting divergence of the harmonic series.

Problem 2

Exercise 6.5.2

Problem 3: Absolute convergence at $c \Rightarrow$ uniform convergence on $[-|c|, |c|]$

Exercise 6.5.3

Theorem 3: If a power series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely at a point x_0, then it converges uniformly on the closed interval $[-c, c]$, where $c = |x_0|$.

Proof: Define

$$M_n = |a_n x_0^n|$$

and let x be any point in $[-c, c]$ where $c = |x_0|$. Then

$$|x| \leq |x_0|$$

so, by the multiplicative property of the absolute value function,

$$|a_n x^n| \leq |a_n x_0^n| = M_n.$$

Absolute convergence of $\sum_{n=0}^{\infty} a_n x_0^n$ is equivalent to saying that the sum $\sum_{n=0}^{\infty} M_n$ converges. By the Weierstrass M-Test, $\sum_{n=0}^{\infty} a_n x^n$ converges uniformly whenever $x \in [-c, c]$.

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Problem 4

Exercise 6.5.5

Problem 5

Exercise 6.5.7

Problem 6

Exercise 6.5.8

Problem 7

Exercise 6.6.3

Problem 8

Exercise 6.6.5, also Exercise 2.8.7 as a prerequisite

Problem 9

Exercise 6.6.7

Problem 10

Exercise 6.6.8

Problem 11
Exercise 6.6.9

Problem 12

Exercise 6.6.10